



GROUP ASSIGNMENT (PART 2)

TECHNOLOGY PARK MALAYSIA

CT042-3-1-IDB

INTRODUCTION TO DATABASES

INTAKE CODE: **APU1F2411CS**

MODULE LECTURER: **LAI CHEW PING**

ASSIGNMENT HAND OUT DATE: **14 JANUARY 2025**

ASSIGNMENT HAND IN DATE: **16 MARCH 2025**

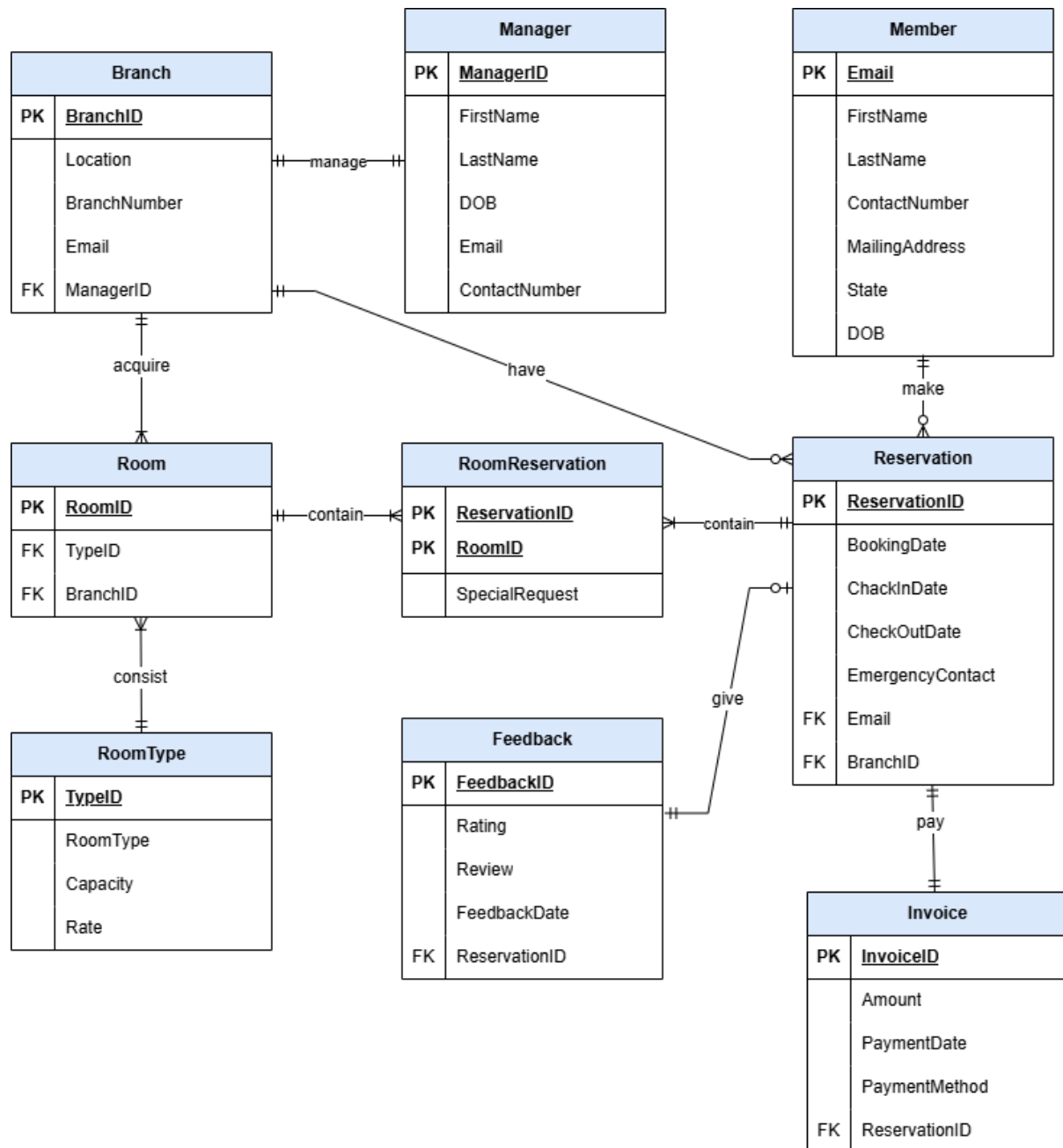
NO.	STUDENT NAME	TP NUMBER
1	SITI NUR ZAFIRAH BINTI ZUNAIID	TP077333
2	NURUN NAJIHAH BINTI A AZMI	TP077342
3	NUR AINA DALILI BINTI MOHD RAFIX	TP077312
4	ALIAH BINTI MOHAMMED FAKHRUL FAHMI	TP077337
5	AMIRAH HUSNA BINTI AZIZAN	TP077336

Table of Contents

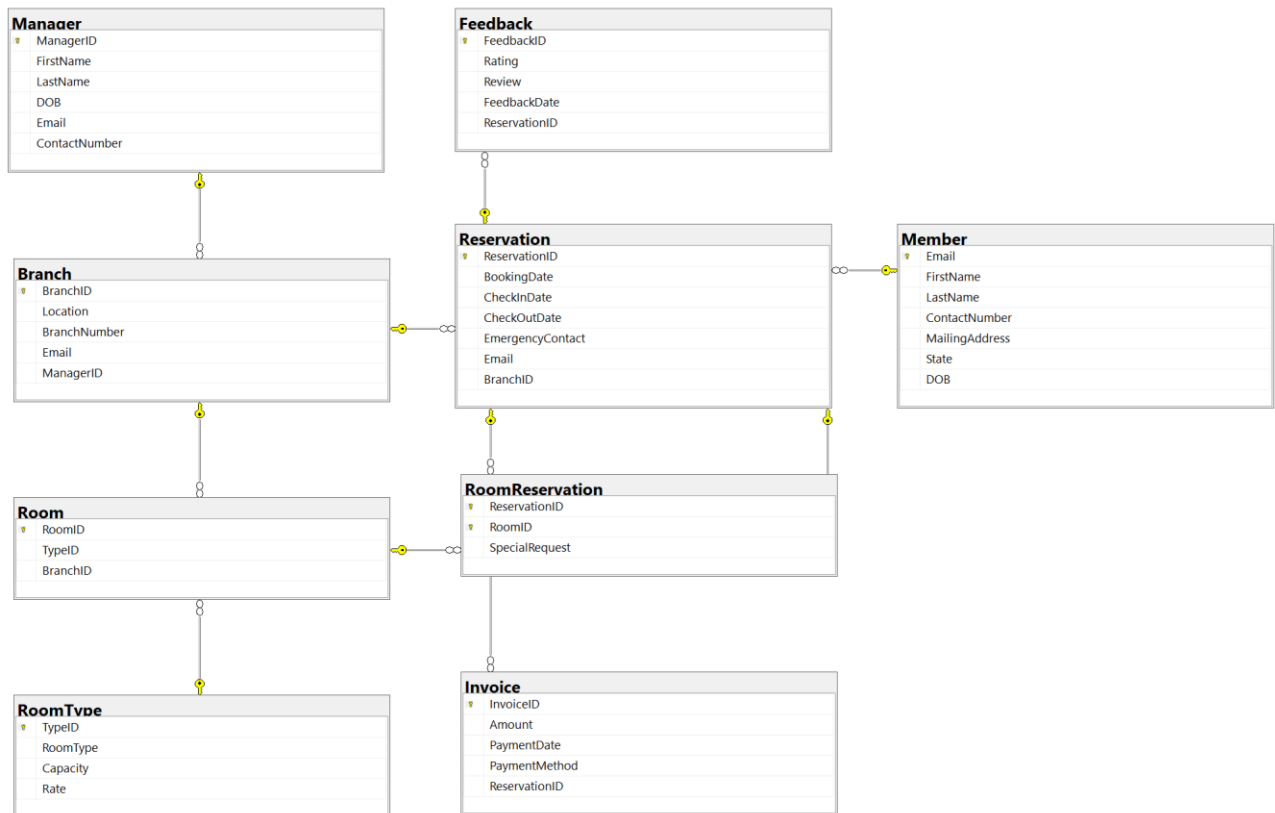
1.0	Database Schema.....	3
1.1	Entity Relationship Diagram.....	3
1.2	Database Diagram	4
2.0	SQL-Data Definition Language (DDL).....	5
3.0	SQL-Data Manipulation Language (DML).....	13
4.0	Workload Matrix	19

1.0 Database Schema

1.1 Entity Relationship Diagram



1.2 Database Diagram



2.0 SQL-Data Definition Language (DDL)

Create and use the database.

```
create database Hotel_ManagementIDB;
```

```
use Hotel_ManagementIDB;
```

Create table and insert data into table.

1. Manager table

Query:

```
Create table Manager
(ManagerID nvarchar(50) Primary Key,
FirstName nvarchar(50) NOT NULL,
LastName nvarchar(50) NOT NULL,
DOB date NOT NULL,
Email nvarchar(50) NOT NULL,
ContactNumber nvarchar(50) NOT NULL);

insert into Manager values
('M001', 'Anis', 'Razak', '2000-04-19', 'anisrazak@mail.com', '012-2768907'),
('M002', 'Abu', 'Bakar', '1995-11-07', 'abubakar123@mail.com', '011-20657893'),
('M003', 'Lee', 'Jeno', '2005-03-01', 'lee.jeno@mail.com', '010-5546788'),
('M004', 'Shawn', 'James', '1999-02-09', 'shawn_mendes@mail.com', '019-5534211');
```

Result:

	ManagerID	FirstName	LastName	DOB	Email	ContactNumber
1	M001	Anis	Razak	2000-04-19	anisrazak@mail.com	012-2768907
2	M002	Abu	Bakar	1995-11-07	abubakar123@mail.com	011-20657893
3	M003	Lee	Jeno	2005-03-01	lee.jeno@mail.com	010-5546788
4	M004	Shawn	James	1999-02-09	shawn_mendes@mail.com	019-5534211

2. Branch table

Query:

```
Create table Branch
(BranchID nvarchar(50) Primary Key,
[Location] nvarchar(50) NOT NULL,
BranchNumber nvarchar(50) NOT NULL,
Email nvarchar(50) NOT NULL,
ManagerID nvarchar(50) NOT NULL Foreign Key References Manager(ManagerID));

insert into Branch values
('X001', 'Kuala Lumpur', '03-6767000', 'xhotelkl@mail.com', 'M001'),
('X002', 'Penang', '03-6781000', 'xhotelpenang@mail.com', 'M002'),
('X003', 'Johor', '03-8090000', 'xhoteljohor@mail.com', 'M003'),
('X004', 'Melaka', '03-4781000', 'xhotelkedah@mail.com', 'M004');
```

Result:

	BranchID	Location	BranchNumber	Email	ManagerID
1	X001	Kuala Lumpur	03-6767000	xhotelkl@mail.com	M001
2	X002	Penang	03-6781000	xhotelpenang@mail.com	M002
3	X003	Johor	03-8090000	xhoteljohor@mail.com	M003
4	X004	Melaka	03-4781000	xhotelkedah@mail.com	M004

3. Member table

Query:

```

Create table [Member]
(Email nvarchar(50) Primary Key,
FirstName nvarchar(50) NOT NULL,
LastName nvarchar(50) NOT NULL,
ContactNumber nvarchar(50) NOT NULL,
MailingAddress nvarchar(50) NOT NULL,
[State] nvarchar(50) NOT NULL,
DOB date NOT NULL);

insert into [Member] values
('johndoe@mail.com', 'John', 'Doe', '013-4567890', '45A Jalan Sentosa', 'Penang', '1997-11-24'),
('alice.smith@mail.com', 'Alice', 'Smith', '017-9876543', '123 Main Street', 'Kuala Lumpur', '1991-07-17'),
('olivia.rodrido@mail.com', 'Olivia', 'Rodrigo', '016-7438892', '789 Taman Indah', 'Johor', '1993-05-06'),
('lewis.capaldi@mail.com', 'Lewis', 'Capaldi', '011-3421906', '56 Lorong Aman', 'Kelantan', '1989-01-10'),
('tan.kim@mail.com', 'Kim', 'Tan', '013-3494647', '10-2A Persiaran Maju', 'Selangor', '1992-02-22'),
('caprice90@mail.com', 'Caprice', 'Lewis', '012-3679087', '674 Jalan Mutiara', 'Sarawak', '1978-11-24'),
('arif.azmi@mail.com', 'Arif', 'Azmi', '019-1907342', '73-2C Persiaran Permata', 'Perak', '2002-01-10'),
('jenniekim@mail.com', 'Jennie', 'Kim', '017-3672545', '34 Taman Sekamat', 'Selangor', '1995-09-18'),
('georgie.steve@mail.com', 'Georgie', 'Steve', '011-61802757', '545 Jalan Aman', 'Negeri Sembilan', '1987-03-17'),
('sofia.car@mail.com', 'Sofia', 'Carter', '012-6480220', '99 Lorong Nenas', 'Selangor', '2001-06-09'),
('ahmadali@mail.com', 'Ahmad', 'Ali', '017-2767650', '90-1B Persiaran Indah', 'Pahang', '1985-04-03'),
('melur@mail.com', 'Melur', 'Azhar', '013-8923346', '12 Jalan Kasih', 'Johor', '1976-08-08'),
('aminah123@mail.com', 'Aminah', 'Bakri', '010-8974530', 'Lot 776 Jalan Merdeka', 'Terengganu', '1999-11-23'),
('theweekdys@mail.com', 'Louise', 'Patridge', '011-98795604', '56 Jalan Jambu', 'Melaka', '1994-12-01'),
('jacobstec@mail.com', 'Jacob', 'Stec', '017-2836754', '245 Persiaran Puteri', 'Sabah', '1994-05-16'),
('arissazul@mail.com', 'Arisa', 'Zul', '012-8974567', '83 Lorong Mawar 2', 'Kedah', '2000-01-31');

```

Result:

	Email	FirstName	LastName	ContactNumber	MailingAddress	State	DOB
1	ahmadali@mail.com	Ahmad	Ali	017-2767650	90-1B Persiaran Indah	Pahang	1985-04-03
2	alice.smith@mail.com	Alice	Smith	017-9876543	123 Main Street	Kuala Lumpur	1991-07-17
3	aminah123@mail.com	Aminah	Bakri	010-8974530	Lot 776 Jalan Merdeka	Terengganu	1999-11-23
4	arif.azmi@mail.com	Arif	Azmi	019-1907342	73-2C Persiaran Permata	Perak	2002-01-10
5	arissazul@mail.com	Arisa	Zul	012-8974567	83 Lorong Mawar 2	Kedah	2000-01-31
6	caprice90@mail.com	Caprice	Lewis	012-3679087	674 Jalan Mutiara	Sarawak	1978-11-24
7	georgie.steve@mail.com	Georgie	Steve	011-61802757	545 Jalan Aman	Negeri Sembilan	1987-03-17
8	jacobstec@mail.com	Jacob	Stec	017-2836754	245 Persiaran Puteri	Sabah	1994-05-16
9	jenniekim@mail.com	Jennie	Kim	017-3672545	34 Taman Sekamat	Selangor	1995-09-18
10	johndoe@mail.com	John	Doe	013-4567890	45A Jalan Sentosa	Penang	1997-11-24
11	lewis.capaldi@mail.com	Lewis	Capaldi	011-3421906	56 Lorong Aman	Kelantan	1989-01-10
12	melur@mail.com	Melur	Azhar	013-8923346	12 Jalan Kasih	Johor	1976-08-08
13	olivia.rodrido@mail.com	Olivia	Rodrigo	016-7438892	789 Taman Indah	Johor	1993-05-06
14	sofia.car@mail.com	Sofia	Carter	012-6480220	99 Lorong Nenas	Selangor	2001-06-09
15	tan.kim@mail.com	Kim	Tan	013-3494647	10-2A Persiaran Maju	Selangor	1992-02-22
16	theweekdys@mail.com	Louise	Patridge	011-98795604	56 Jalan Jambu	Melaka	1994-12-01

4. RoomType table

Query:

```
Create table RoomType
(TypeID nvarchar(50) Primary Key,
RoomType nvarchar(50) NOT NULL,
Capacity int NOT NULL,
Rate decimal(10,2) NOT NULL);

insert into RoomType values
('T001', 'Standard Double', 2, 200.00),
('T002', 'Standard Twin', 2, 180.00),
('T003', 'Deluxe Double', 4, 300.00);
```

Result:

	TypeID	RoomType	Capacity	Rate
1	T001	Standard Double	2	200.00
2	T002	Standard Twin	2	180.00
3	T003	Deluxe Double	4	300.00

5. Room table

Query:

```
Create table Room
(RoomID nvarchar(50) Primary Key,
TypeID nvarchar(50) NOT NULL Foreign Key References RoomType(TypeID),
BranchID nvarchar(50) NOT NULL Foreign Key References Branch(BranchID));

insert into Room values
('KL001', 'T001', 'X001'),
('KL002', 'T001', 'X001'),
('KL003', 'T002', 'X001'),
('KL004', 'T002', 'X001'),
('KL005', 'T003', 'X001'),
('KL006', 'T003', 'X001'),
('P001', 'T001', 'X002'),
('P002', 'T001', 'X002'),
('P003', 'T002', 'X002'),
('P004', 'T002', 'X002'),
('P005', 'T003', 'X002'),
('P006', 'T003', 'X002'),
('J001', 'T001', 'X003'),
('J002', 'T001', 'X003'),
('J003', 'T002', 'X003'),
('J004', 'T002', 'X003'),
('J005', 'T003', 'X003'),
('J006', 'T003', 'X003'),
('ML001', 'T001', 'X004'),
('ML002', 'T001', 'X004'),
('ML003', 'T002', 'X004'),
('ML004', 'T002', 'X004'),
('ML005', 'T003', 'X004'),
('ML006', 'T003', 'X004');
```

Result:

	RoomID	TypeID	BranchID
1	J001	T001	X003
2	J002	T001	X003
3	J003	T002	X003
4	J004	T002	X003
5	J005	T003	X003
6	J006	T003	X003
7	KL001	T001	X001
8	KL002	T001	X001
9	KL003	T002	X001
10	KL004	T002	X001
11	KL005	T003	X001
12	KL006	T003	X001
13	ML001	T001	X004
14	ML002	T001	X004
15	ML003	T002	X004
16	ML004	T002	X004
17	ML005	T003	X004
18	ML006	T003	X004
19	P001	T001	X002
20	P002	T001	X002
21	P003	T002	X002
22	P004	T002	X002
23	P005	T003	X002
24	P006	T003	X002

6. Reservation table

Query:

```
Create table Reservation
(ReservationID nvarchar(50) Primary Key,
BookingDate date NOT NULL,
CheckInDate date NOT NULL,
CheckOutDate date NOT NULL,
EmergencyContact nvarchar(50) NOT NULL,
Email nvarchar(50) NOT NULL Foreign Key References [Member](Email),
BranchID nvarchar(50) NOT NULL Foreign Key References Branch(BranchID));
```



```

insert into Reservation values
('R001', '2023-09-01', '2023-09-10', '2023-09-14', '012-3456789', 'johndoe@mail.com', 'X001'),
('R002', '2023-12-12', '2024-01-01', '2024-01-03', '010-877-5606', 'olivia.rodrido@mail.com', 'X001'),
('R003', '2024-01-05', '2024-01-15', '2024-01-17', '013-3423143', 'jenniekim@mail.com', 'X002'),
('R004', '2024-01-05', '2024-01-19', '2024-01-22', '013-3423143', 'jenniekim@mail.com', 'X004'),
('R005', '2024-01-31', '2024-02-04', '2024-02-07', '017-9876543', 'alice.smith@mail.com', 'X003'),
('R006', '2024-02-08', '2024-02-16', '2024-02-19', '011-3458873', 'caprice90@mail.com', 'X001'),
('R007', '2024-02-15', '2024-02-25', '2024-03-01', '010-1122345', 'sofia.car@mail.com', 'X002'),
('R008', '2024-03-10', '2024-03-13', '2024-03-16', '013-3423143', 'jenniekim@mail.com', 'X004'),
('R009', '2024-03-10', '2024-03-13', '2024-03-17', '019-9907845', 'melur@mail.com', 'X004'),
('R010', '2024-03-10', '2024-04-15', '2024-04-18', '019-9907845', 'melur@mail.com', 'X003'),
('R011', '2024-06-05', '2024-06-20', '2024-06-24', '011-9807812', 'georgie.steve@mail.com', 'X001'),
('R012', '2024-07-19', '2024-07-25', '2024-07-28', '019-345-6538', 'lewis.capldi@mail.com', 'X002'),
('R013', '2024-08-03', '2024-08-09', '2024-08-11', '010-877-5606', 'olivia.rodrido@mail.com', 'X001'),
('R014', '2024-10-01', '2024-11-01', '2024-11-05', '019-9907845', 'melur@mail.com', 'X001'),
('R015', '2024-11-25', '2024-12-25', '2024-12-29', '010-1234987', 'aminah123@mail.com', 'X001'),
('R016', '2024-11-27', '2024-12-05', '2024-12-10', '011-3451231', 'ahmadali@mail.com', 'X004'),
('R017', '2024-12-18', '2024-12-25', '2024-12-28', '013-3423143', 'jenniekim@mail.com', 'X002'),
('R018', '2024-12-18', '2024-12-25', '2024-12-29', '019-9907845', 'melur@mail.com', 'X001'),
('R019', '2025-01-15', '2025-02-01', '2025-02-05', '013-3423143', 'jenniekim@mail.com', 'X001'),
('R020', '2025-02-14', '2025-03-01', '2025-03-04', '012-6754331', 'theweekdys@mail.com', 'X003'),
('R021', '2025-02-20', '2025-04-26', '2025-05-01', '013-3494647', 'tan.kim@mail.com', 'X004'),
('R022', '2025-02-20', '2025-06-07', '2025-06-10', '010-1234987', 'aminah123@mail.com', 'X002'),
('R023', '2025-03-13', '2025-08-19', '2025-08-23', '013-3423143', 'jenniekim@mail.com', 'X004');

```

Result:

	ReservationID	BookingDate	CheckInDate	CheckOutDate	EmergencyContact	Email	BranchID
1	R001	2023-09-01	2023-09-10	2023-09-14	012-3456789	johndoe@mail.com	X001
2	R002	2023-12-12	2024-01-01	2024-01-03	010-877-5606	olivia.rodrido@mail.com	X001
3	R003	2024-01-05	2024-01-15	2024-01-17	013-3423143	jenniekim@mail.com	X002
4	R004	2024-01-05	2024-01-19	2024-01-22	013-3423143	jenniekim@mail.com	X004
5	R005	2024-01-31	2024-02-04	2024-02-07	017-9876543	alice.smith@mail.com	X003
6	R006	2024-02-08	2024-02-16	2024-02-19	011-3458873	caprice90@mail.com	X001
7	R007	2024-02-15	2024-02-25	2024-03-01	010-1122345	sofia.car@mail.com	X002
8	R008	2024-03-10	2024-03-13	2024-03-16	013-3423143	jenniekim@mail.com	X004
9	R009	2024-03-10	2024-03-13	2024-03-17	019-9907845	melur@mail.com	X004
10	R010	2024-03-10	2024-04-15	2024-04-18	019-9907845	melur@mail.com	X003
11	R011	2024-06-05	2024-06-20	2024-06-24	011-9807812	georgie.steve@mail.com	X001
12	R012	2024-07-19	2024-07-25	2024-07-28	019-345-6538	lewis.capldi@mail.com	X002
13	R013	2024-08-03	2024-08-09	2024-08-11	010-877-5606	olivia.rodrido@mail.com	X001
14	R014	2024-10-01	2024-11-01	2024-11-05	019-9907845	melur@mail.com	X001
15	R015	2024-11-25	2024-12-25	2024-12-29	010-1234987	aminah123@mail.com	X001
16	R016	2024-11-27	2024-12-05	2024-12-10	011-3451231	ahmadali@mail.com	X004
17	R017	2024-12-18	2024-12-25	2024-12-28	013-3423143	jenniekim@mail.com	X002
18	R018	2024-12-18	2024-12-25	2024-12-29	019-9907845	melur@mail.com	X001
19	R019	2025-01-15	2025-02-01	2025-02-05	013-3423143	jenniekim@mail.com	X001
20	R020	2025-02-14	2025-03-01	2025-03-04	012-6754331	theweekdys@mail.com	X003
21	R021	2025-02-20	2025-04-26	2025-05-01	013-3494647	tan.kim@mail.com	X004
22	R022	2025-02-20	2025-06-07	2025-06-10	010-1234987	aminah123@mail.com	X002
23	R023	2025-03-13	2025-08-19	2025-08-23	013-3423143	jenniekim@mail.com	X004

7. RoomReservation table

Query:

```

Create table RoomReservation
(ReservationID nvarchar(50),
RoomID nvarchar(50),
SpecialRequest nvarchar(50),
Primary Key (ReservationID, RoomID),
Foreign Key (ReservationID) References Reservation(ReservationID),
Foreign Key (RoomID) References Room(RoomID));

```

```

insert into RoomReservation values
('R001', 'KL001', NULL),
('R001', 'KL005', NULL),
('R002', 'KL001', NULL),
('R003', 'P003', 'Non-smoking'),
('R004', 'ML005', NULL),
('R005', 'J005', NULL),
('R006', 'KL001', 'Baby cot'),
('R006', 'KL002', 'Extra pillow'),
('R006', 'KL005', 'Late check-in'),
('R007', 'J001', NULL),
('R008', 'ML006', NULL),
('R009', 'ML004', 'Extra towels'),
('R009', 'ML006', 'Birthday cake'),
('R010', 'J001', NULL),
('R011', 'KL002', NULL),
('R012', 'P002', 'Netflix'),
('R013', 'ML004', 'Hypoallergenic pillow'),
('R014', 'KL002', NULL),
('R014', 'KL003', 'Extra towels'),
('R015', 'KL002', NULL),
('R016', 'ML001', NULL),
('R017', 'P002', 'Extra toiletries'),
('R018', 'KL004', 'With KLCC view'),
('R018', 'KL006', NULL),
('R019', 'KL004', NULL),
('R020', 'J006', 'Anniversary decorations'),
('R021', 'ML004', NULL),
('R022', 'P004', NULL),
('R023', 'ML005', NULL);

```

Result:

	ReservationID	RoomID	SpecialRequest
1	R001	KL001	NULL
2	R001	KL005	NULL
3	R002	KL001	NULL
4	R003	P003	Non-smoking
5	R004	ML005	NULL
6	R005	J005	NULL
7	R006	KL001	Baby cot
8	R006	KL002	Extra pillow
9	R006	KL005	Late check-in
10	R007	J001	NULL
11	R008	ML006	NULL
12	R009	ML004	Extra towels
13	R009	ML006	Birthday cake
14	R010	J001	NULL
15	R011	KL002	NULL
16	R012	P002	Netflix
17	R013	ML004	Hypoallergenic pillow
18	R014	KL002	NULL
19	R014	KL003	Extra towels
20	R015	KL002	NULL
21	R016	ML001	NULL
22	R017	P002	Extra toiletries
23	R018	KL004	With KLCC view
24	R018	KL006	NULL
25	R019	KL004	NULL
26	R020	J006	Anniversary decora...
27	R021	ML004	NULL
28	R022	P004	NULL
29	R023	ML005	NULL

8. Invoice table

Query:

```

Create table Invoice
(InvoiceID nvarchar(50) Primary Key,
Amount decimal(10,2) NOT NULL,
PaymentDate date NOT NULL,
PaymentMethod nvarchar(50) NOT NULL,
ReservationID nvarchar(50) NOT NULL Foreign Key References Reservation(ReservationID));

insert into Invoice values
('INV001', 2000.00, '2023-09-01', 'Cash', 'R001'),
('INV002', 400.00, '2023-12-12', 'Online', 'R002'),
('INV003', 360.00, '2024-01-05', 'Cash', 'R003'),
('INV004', 900.00, '2024-01-05', 'Online', 'R004'),
('INV005', 900.00, '2024-01-31', 'Online', 'R005'),
('INV006', 2100.00, '2024-02-08', 'Online', 'R006'),
('INV007', 1000.00, '2024-02-15', 'Online', 'R007'),
('INV008', 900.00, '2024-03-10', 'Cash', 'R008'),
('INV009', 1920.00, '2024-03-10', 'Online', 'R009'),
('INV010', 600.00, '2024-03-10', 'Online', 'R010'),
('INV011', 800.00, '2024-06-05', 'Cash', 'R011'),
('INV012', 600.00, '2024-07-19', 'Online', 'R012'),
('INV013', 360.00, '2024-08-03', 'Online', 'R013'),
('INV014', 1520.00, '2024-10-01', 'Cash', 'R014'),
('INV015', 800.00, '2024-11-25', 'Online', 'R015'),
('INV016', 1000.00, '2024-11-27', 'Online', 'R016'),
('INV017', 600.00, '2024-12-18', 'Cash', 'R017'),
('INV018', 1920.00, '2024-12-18', 'Online', 'R018'),
('INV019', 720.00, '2025-01-15', 'Cash', 'R019'),
('INV020', 900.00, '2025-02-14', 'Online', 'R020'),
('INV021', 1500.00, '2025-02-20', 'Online', 'R021'),
('INV022', 540.00, '2025-02-20', 'Online', 'R022'),
('INV023', 1200.00, '2025-03-13', 'Online', 'R023');

```

Result:

	InvoiceID	Amount	PaymentDate	PaymentMethod	ReservationID
1	INV001	2000.00	2023-09-01	Cash	R001
2	INV002	400.00	2023-12-12	Online	R002
3	INV003	360.00	2024-01-05	Cash	R003
4	INV004	900.00	2024-01-05	Online	R004
5	INV005	900.00	2024-01-31	Online	R005
6	INV006	2100.00	2024-02-08	Online	R006
7	INV007	1000.00	2024-02-15	Online	R007
8	INV008	900.00	2024-03-10	Cash	R008
9	INV009	1920.00	2024-03-10	Online	R009
10	INV010	600.00	2024-03-10	Online	R010
11	INV011	800.00	2024-06-05	Cash	R011
12	INV012	600.00	2024-07-19	Online	R012
13	INV013	360.00	2024-08-03	Online	R013
14	INV014	1520.00	2024-10-01	Cash	R014
15	INV015	800.00	2024-11-25	Online	R015
16	INV016	1000.00	2024-11-27	Online	R016
17	INV017	600.00	2024-12-18	Cash	R017
18	INV018	1920.00	2024-12-18	Online	R018
19	INV019	720.00	2025-01-15	Cash	R019
20	INV020	900.00	2025-02-14	Online	R020
21	INV021	1500.00	2025-02-20	Online	R021
22	INV022	540.00	2025-02-20	Online	R022
23	INV023	1200.00	2025-03-13	Online	R023

9. Feedback table

Query:

```

Create table Feedback
(FeedbackID nvarchar(50) Primary Key,
Rating int NOT NULL check (Rating BETWEEN 1 AND 5),
Review nvarchar(50),
FeedbackDate date NOT NULL,
ReservationID nvarchar(50) NOT NULL Foreign Key References Reservation(ReservationID));

insert into Feedback values
('F001', 4, 'Satisfied', '2023-09-01', 'R001'),
('F002', 5, 'Nice very comfy', '2023-12-12', 'R002'),
('F003', 3, 'Hotel is nice but did not get non-smoking room', '2024-01-07', 'R003'),
('F004', 5, 'Excellent services. Good job', '2024-01-05', 'R004'),
('F005', 2, 'Construction is too loud', '2024-01-31', 'R005'),
('F006', 5, NULL, '2024-02-13', 'R006'),
('F009', 5, 'The cake tastes divine', '2024-03-10', 'R009'),
('F010', 3, 'Stuff took so long to come to my room', '2024-03-11', 'R010'),
('F012', 5, 'Staff was very professional', '2024-07-19', 'R012'),
('F014', 5, 'Mesmerizing view of KL from our room', '2024-10-01', 'R014'),
('F015', 4, 'The room is very accessible', '2024-11-26', 'R015'),
('F018', 5, 'The view of KLCC Twin Tower is so mesmerising', '2024-12-18', 'R018'),
('F019', 2, 'Stuff is not welcoming', '2025-01-16', 'R019'),
('F020', 5, 'The decorations are very nice', '2025-02-14', 'R020'),
('F022', 5, NULL, '2025-02-22', 'R022');

```

Result:

	FeedbackID	Rating	Review	FeedbackDate	ReservationID
1	F001	4	Satisfied	2023-09-01	R001
2	F002	5	Nice very comfy	2023-12-12	R002
3	F003	3	Hotel is nice but did not get non-smoking room	2024-01-07	R003
4	F004	5	Excellent services. Good job	2024-01-05	R004
5	F005	2	Construction is too loud	2024-01-31	R005
6	F006	5	NULL	2024-02-13	R006
7	F009	5	The cake tastes divine	2024-03-10	R009
8	F010	3	Stuff took so long to come to my room	2024-03-11	R010
9	F012	5	Staff was very professional	2024-07-19	R012
10	F014	5	Mesmerizing view of KL from our room	2024-10-01	R014
11	F015	4	The room is very accessible	2024-11-26	R015
12	F018	5	The view of KLCC Twin Tower is so mesmerising	2024-12-18	R018
13	F019	2	Stuff is not welcoming	2025-01-16	R019
14	F020	5	The decorations are very nice	2025-02-14	R020
15	F022	5	NULL	2025-02-22	R022

3.0 SQL-Data Manipulation Language (DML)

- i. Show the name, contact number, and email address of every branch manager.

Query:

```
Select FirstName, LastName, Email, ContactNumber from Manager;
```

Result:

	FirstName	LastName	Email	ContactNumber
1	Anis	Razak	anisrazak@mail.com	012-2768907
2	Abu	Bakar	abubakar123@mail.com	011-20657893
3	Lee	Jeno	lee.jeno@mail.com	010-5546788
4	Shawn	James	shawn_mendes@mail.com	019-5534211

- ii. Show the first name and last name of the customers who had not made any reservation in 2024.

Query:

```
Select FirstName, LastName, Reservation.BookingDate from [Member]
full outer Join Reservation ON Member.Email = Reservation.Email
Where Member.Email not in (Select Email from Reservation)
OR Member.Email not in (Select Email from Reservation where year(BookingDate) = '2024');
```

Result:

	FirstName	LastName	BookingDate
1	Arif	Azmi	NULL
2	Arisa	Zul	NULL
3	Jacob	Stec	NULL
4	John	Doe	2023-09-01
5	Kim	Tan	2025-02-20
6	Louise	Patridge	2025-02-14

- iii. Show the customers with the highest number of reservations in 2024. The result should contain first name, last name, email address, and contact number.

Query:

```
Select top 1 Member.FirstName, Member.LastName, Member.Email, Member.ContactNumber from Member
join Reservation on Member.Email = Reservation.Email
Where Year(Reservation.BookingDate) = 2024;
```

Result:

	FirstName	LastName	Email	ContactNumber
1	Jennie	Kim	jenniekim@mail.com	017-3672545

- iv. Show the total and average revenue generated by each branch in 2024.

Query:

```
Select Branch.[Location], SUM(Invoice.Amount) as TotalRevenue, AVG(Invoice.Amount) as AverageRevenue
From Reservation
Join Branch ON Branch.BranchID = Reservation.BranchID
Join Invoice ON Invoice.ReservationID = Reservation.ReservationID
Where Reservation.BookingDate between '2024/01/01' AND '2024/12/31'
Group by Branch.[Location];
```

Result:

	Location	TotalRevenue	AverageRevenue
1	Johor	1500.00	750.000000
2	Kuala Lumpur	7500.00	1250.000000
3	Melaka	4720.00	1180.000000
4	Penang	2560.00	640.000000

- v. Show the average, maximum, minimum, and total number of ratings by each branch.

Query:

```
Select b.[Location], AVG(f.Rating) as AvgRating, MAX(f.Rating) as MaxRating,
MIN(f.Rating) as MinRating, COUNT(f.Rating) as TotalRatings
from Reservation r
Join Branch b ON b.BranchID = r.BranchID
Join Feedback f ON f.ReservationID = r.ReservationID
Group by b.[Location];
```

Result:

	Location	AvgRating	MaxRating	MinRating	TotalRatings
1	Johor	3	5	2	3
2	Kuala Lumpur	4	5	2	7
3	Melaka	5	5	5	2
4	Penang	4	5	3	3

- vi. Show the total number of rooms for each type (Standard Double, Standard Twin, and Deluxe Double) in each branch.

Query:

```
Select Branch.Location, RoomType.RoomType, COUNT(Room.RoomID) as TotalRooms
from Room
Join Branch on Branch.BranchID = Room.BranchID
Join RoomType on RoomType.TypeID = Room.TypeID
Group by RoomType.RoomType, Branch.Location;
```

Result:

	Location	RoomType	TotalRooms
1	Johor	Deluxe Double	2
2	Johor	Standard Double	2
3	Johor	Standard Twin	2
4	Kuala Lumpur	Deluxe Double	2
5	Kuala Lumpur	Standard Double	2
6	Kuala Lumpur	Standard Twin	2
7	Melaka	Deluxe Double	2
8	Melaka	Standard Double	2
9	Melaka	Standard Twin	2
10	Penang	Deluxe Double	2
11	Penang	Standard Double	2
12	Penang	Standard Twin	2

- vii. Show the monthly revenue for Kuala Lumpur branch in 2024 and sort the results by month in ascending order.

Query:

```
Select Month(PaymentDate) as Month, SUM(Amount) as MonthlyRevenue from Invoice
Join Reservation ON Invoice.ReservationID = Reservation.ReservationID
Where Reservation.BranchID = 'X001' AND YEAR(Invoice.PaymentDate) = 2024
Group by Month(Invoice.PaymentDate)
Order by [Month] ASC
```

Result:

	Month	MonthlyRevenue
1	2	2100.00
2	6	800.00
3	8	360.00
4	10	1520.00
5	11	800.00
6	12	1920.00

- viii. Show the total number of room types of each branch.

Query:

```
Select Branch.[Location] as Branch, COUNT(DISTINCT(TypeID)) AS TotalRoomTypes
From Room
Join Branch ON Branch.BranchID = Room.BranchID
Group by Branch.[Location];
```

Result:

	Branch	TotalRoomTypes
1	Johor	3
2	Kuala Lumpur	3
3	Melaka	3
4	Penang	3

- ix. Show the average rating by room type and sort the result by average rating in descending order.

Query:

```
Select rt.RoomType, AVG(f.Rating) as AvgRating
from Feedback f
Join RoomReservation rr ON rr.ReservationID = f.ReservationID
Join Room r ON r.RoomID = rr.RoomID
Join RoomType rt on rt.TypeID = r.TypeID
Group by rt.RoomType
Order by AvgRating DESC;
```

Result:

	RoomType	AvgRating
1	Deluxe Double	4
2	Standard Double	4
3	Standard Twin	4

- x. Show the number of reservations made by each customer whose last name begins with the letter "A".

Query:

```
Select Member.FirstName, Member.LastName, Count(*) as NumberReservation
From Reservation
Join Member ON Member.Email = Reservation.Email
Where Member.LastName LIKE 'A%'
Group by Member.FirstName, Member.LastName
```

Result:

	FirstName	LastName	NumberReservation
1	Ahmad	Ali	1
2	Melur	Azhar	4

- xi. Show all the upcoming bookings for the Melaka branch. The result should include the customer's first and last name, check-in date, check-out date, and room type.

Query:

```
Select m.FirstName, m.LastName, r.CheckInDate, r.CheckOutDate, rt.RoomType
From Reservation r
Join Member m ON m.Email = r.Email
Join RoomReservation rr ON rr.ReservationID = r.ReservationID
Join Room ro ON ro.RoomID = rr.RoomID
Join RoomType rt ON rt.TypeID = ro.TypeID
Where r.BranchID = 'X004' AND r.CheckInDate > GETDATE();
```

Result:

	FirstName	LastName	CheckInDate	CheckOutDate	RoomType
1	Kim	Tan	2025-04-26	2025-05-01	Deluxe Double
2	Jennie	Kim	2025-08-19	2025-08-23	Deluxe Double

- xii. Show the reservation with multiple rooms. The result should include reservation id, customer's first and last name, along with the total number of rooms booked.

Query:

```
Select r.ReservationID, m.FirstName, m.LastName, COUNT(rr.RoomID) AS TotalRoomsBooked
From Reservation r
Join Member m ON m.Email = r.Email
Join RoomReservation rr ON rr.ReservationID = r.ReservationID
Group by r.ReservationID, m.FirstName, m.LastName
Having COUNT(rr.RoomID) > 1;
```

Result:

	ReservationID	FirstName	LastName	TotalRoomsBooked
1	R001	John	Doe	2
2	R006	Caprice	Lewis	3
3	R009	Melur	Azhar	2
4	R014	Melur	Azhar	2
5	R018	Melur	Azhar	2

- xiii. Show the customer who had made more than 3 reservations in 2024. The result should include first name, last name, email address, and contact number of the customer.

Query:

```
Select FirstName, LastName, Email, ContactNumber
From Member
Where Email IN
(Select Email from Reservation
Where BookingDate between '2024/01/01' AND '2024/12/31'
Group by Email
Having COUNT(ReservationID) > 3);
```

Result:

	FirstName	LastName	Email	ContactNumber
1	Jennie	Kim	jenniekim@mail.com	017-3672545
2	Melur	Azhar	melur@mail.com	013-8923346

- xiv. Show all reservations with a check-in date of 2024-12-25. Include the customer's first and last names, check-in date, and check-out date in the results.

Query:

```
Select m.FirstName, m.LastName, r.CheckInDate, r.CheckOutDate
From Reservation r
Join Member m ON r.Email = m.Email
Where r.CheckInDate = '2024-12-25';
```

Result:

	FirstName	LastName	CheckInDate	CheckOutDate
1	Aminah	Bakri	2024-12-25	2024-12-29
2	Jennie	Kim	2024-12-25	2024-12-28
3	Melur	Azhar	2024-12-25	2024-12-29

- xv. Show the reservations with ratings below 3, including the feedback message, the customer's email address and the feedback date.

Query:

```
Select r.ReservationID, m.Email, f.Rating, f.Review, f.FeedbackDate
From Reservation r
Join Feedback f ON f.ReservationID = r.ReservationID
Join Member m ON m.Email = r.Email
Where f.Rating < 3;
```

Result:

	ReservationID	Email	Rating	Review	FeedbackDate
1	R005	alice.smith@mail.com	2	Construction is too loud	2024-01-31
2	R019	jenniekim@mail.com	2	Stuff is not welcoming	2025-01-16

4.0 Workload Matrix

Part	Component	Student 1: Siti Nur Zafirah	Student 2: Nur Aina Dalili	Student 3: Nurun Najihah	Student 4: Aliah	Student 5: Amirah Husna	Total
1	a. Database and Database Management System	20%	20%	20%	20%	20%	100%
1	b. Business Rules & Normalization	50%	20%	10%	10%	10%	100%
1	c. Entity Relationship Diagram	35%	12%	12%	30%	11%	100%

Part	Component	Student 1: Siti Nur Zafirah	Student 2: Nur Aina Dalili	Student 3: Nurun Najihah	Student 4: Aliah	Student 5: Amirah Husna	Total
2	d. Database Schema	20%	20%	20%	20%	20%	100%
2	e. DDL	20%	20%	20%	20%	20%	100%
2	f. DML	30%	28%	28%	7%	7%	100%